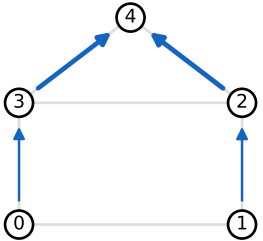
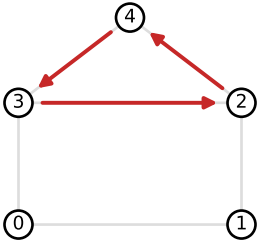
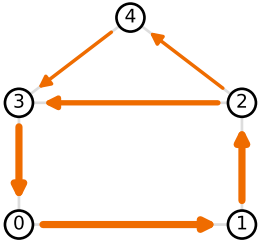
gradient: a ranking



curl: rock-paper-scissors



harmonic: a cycle round a hole